
title: "When Does Guided Search Help?"

subtitle: "A Trustworthy LLM-Agent Autoresearch Protocol for Noisy Engineering Domains, with a Pre-Loop Headroom Diagnostic"

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Abstract

LLM agents for autonomous discovery (FunSearch, AlphaEvolve, AI Scientist) assume exact, verifiable evaluators — a condition that fails in most engineering domains, where data is noisy and no ground truth exists. Independent evaluation documents the cost: 42% of AI Scientist experiments failed on coding errors, 57% of manuscripts contained hallucinated numbers, and published prior art was reported as novel. We present a **trustworthy autoresearch protocol** for such domains, instantiated on real DJI M100 UAV energy modeling and energy-aware path planning, plus a **pre-loop headroom diagnostic** that predicts, before any budget is spent, whether guided search will beat random. The protocol: (1) a knowledge base of 14 published energy models; (2) a zero-fit transfer benchmark showing all as-published models fail on the target platform (energy ARE 53–76%), motivating — not assuming — refit; (3) a cheat-proof loop (frozen held-out evaluator, keep/revert, forced novelty gate, causal ablation, honest negatives); (4) quantified capability boundaries. The diagnostic correctly predicts the energy domain is saturated (tie at 9 controlled complexity levels) and attributes the planning win ($z=-2.38$, 0% of random runs beat it) to cross-space escape into program space. Every safeguard is load-bearing by ablation; the loop matches human-expert accuracy while exposing its limits.

1. Introduction

A delivery drone's flight plan is only as good as its energy model. Under-estimate the cost of climbing over a wall and the planner sends the drone up the expensive route; over-estimate it and the drone wastes battery on a needless detour. Yet the energy model every planner actually runs on — momentum/BEMT theory — is a textbook idealization that has never seen this drone's data. Getting the model right, on real flight measurements, is what determines whether an energy-aware planner saves battery or burns it. And this is not a one-off: every noisy engineering domain (UAV energy, chemical sensor traces, robotic manipulation on hardware) has the same shape — a measurement, not a ground truth; a model that must generalize; an evaluator that can be gamed.

LLM-agent "autoresearch" systems propose, evaluate, and select candidate discoveries in a closed loop. Their appeal is that an LLM can propose in **program space** — new code, new function forms, new solver structures — that classical search (random, Bayesian, evolutionary-with-fixed-mutation) cannot reach. FunSearch discovered a faster matrix-multiplication algorithm; AlphaEvolve improved infrastructure scheduling; Eureka wrote better RL reward functions.

But all of these assume the **evaluator is exact**: FunSearch's is a mathematical proof-checker, AlphaEvolve's a simulation with a known optimum. In real engineering domains — a drone's measured power, a chemical process's sensor traces — there is no exact evaluator. The ground truth is a noisy measurement; the "discovery" is a model that must generalize to unseen operating conditions; and a naive loop can overfit the search split, game a mutable evaluator, or report a known effect as novel.

The contribution of this paper is not a better energy model or a better planner. It is a **protocol** for doing autoresearch honestly in noisy domains, plus a **diagnostic** that tells you up front whether the loop is worth running. We instantiate it on a real, measurable problem — UAV energy modeling and energy-aware path planning on DJI M100 flight data — and we report, deliberately, the settings where the loop **does not** beat random.

Contributions.

1. **Empirical**: zero-shot transfer of 14 published UAV energy models onto a target platform fails systematically (energy ARE 53–76%), motivating an explicit refit stage — measured, not assumed.
2. **Method**: a four-stage trustworthy autoresearch protocol (knowledge base → zero-fit benchmark → cheat-proof loop → boundary quantification) for domains without ground

truth.

3. **Algorithmic: the Headroom Diagnostic**, a cheap pre-loop predictor of whether guided search will beat random, from single-term utility spread and combination synergy. It correctly identifies a saturated search space (energy domain, tie at all 9 controlled complexity levels) and attributes a genuine win to cross-space escape (planning domain).
 4. **Empirical capability law**: guided-search advantage depends on **effective combinatorial complexity**, not nominal space size — ties random in low-effective-dimensional energy (even under nominal-space inflation), beats random in code-structure planning.
 5. **Safeguards are load-bearing**: ablating any one of the protocol's mechanisms makes the loop overclaim or collapse.
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2. Related Work

LLM autoresearch. FunSearch [Romera-Paredes et al., 2024], AlphaEvolve [Novikov et al., 2025], Eureka [Ma et al., 2024], AI Scientist [Lu et al., 2024]. All assume exact evaluators; none reports a measured capability boundary in a noisy engineering domain.

The trustworthiness gap. Independent evaluation of the AI Scientist pipeline [Beel, Kan & Baumgart, ACM SIGIR Forum 2025, arXiv:2502.14297] found 42% experiment coding failures, 57% manuscripts with hallucinated numbers, and keyword-based novelty checks misclassifying established concepts as novel. Recent 2026 evidence sharpens this: the Corral benchmark [Jablonka et al., 2026, 25k+ agent runs across 8 scientific domains] found agents frequently fail to weigh evidence, generate/test hypotheses, or revise assumptions on contradictory results; a position paper ("Falsify, Don't Just Discover") argues AI-generated discoveries are "not born scientific" without automated falsification; a January 2026 case study of four end-to-end autonomous research attempts [arXiv:2601.03315] catalogs six recurring failure modes (bias to training-data defaults, implementation drift, memory degradation, *overexcitement* — declaring success despite failures, weak scientific taste). A *Nature Machine Intelligence* editorial (2026) explicitly recommends that AI-science systems provide ablation analyses and baseline comparisons and transparent reporting — precisely what our protocol delivers. Falsification-first framing is prior art [Liu et al., 2024, AIGS]; our contribution is the engineered protocol plus empirical boundaries. This gap is unoccupied.

Adjacent LLM-drone work. LAENet [arXiv:2505.21045] uses an LLM to design a UAV reward function (Eureka-style, single-shot); LLM-VD couples LLM generation with evolution for truck-drone *routing* (discrete combinatorial VRP). Neither is a closed-loop autoresearch loop over continuous energy *modeling* + planning cost. We cite and differentiate, not claim priority.

UAV energy models. Momentum/BEMT theory (textbook), Tseng regression [Muli, Park & Liu, arXiv:2206.01609], Abeywardena VRS [arXiv:2209.04128], Dorling 2017, Stolaroff 2018, Morbidi 2020. All 14 are in our knowledge base; their forms and sources are in ``energy_model_zoo.py``.

Search-space scaling. Bergstra & Bengio 2012 (random matches/beats grid in low-effective-dimension HPO spaces), Wang et al. 2013 REMBO (~15–20-dim BO tipping point), NAS confounder removal (guided samplers beat random once weight-sharing confounder is removed; ENAS P(surpass random) 0.07→0.90), FunSearch program spaces. Our controlled negative control measures the *nominal-space-inflation does not help* regime directly.

Energy-aware planning (prior art, MUST cite, DON'T claim). EcoFlight 2025; Michel 2024 (delivery order); Nguyen & Au 2017 (reachable set). Our planning result is an *application* of a real-data cost, not a new planning algorithm.

3. Problem Setup

Data. DJI Matrice 100, 209 real flights [Rodrigues et al., Scientific Data 2021], on-board power $P = V \cdot |I|$, payloads 0/250/500g, altitude 25–100 m, speeds 4–12 m/s. Instantaneous power has large non-kinematic variance (kinematics explain <40% of per-sample variance); per-flight *total energy* is the stable quantity.

Frozen evaluator. ``m100_eval.py``. Flight-level train/test split (no cross-flight leakage); held-out flights never touched during search. Metric: per-flight energy ARE (absolute relative error of predicted vs measured total energy), averaged over held-out flights. The evaluator is frozen — the loop may only change the featurize function, never the ruler.

The task. Given the evaluator, find a featurize function $s \mapsto X(s) \in \mathbb{R}^{N \times K}$ whose ridge-fit (trained on the search split) achieves the lowest held-out energy ARE.

4. Method: The Protocol

4.1 Stage 1 — Prior-art knowledge base

Collect published UAV energy model **forms** (14 in ``energy_model_zoo.py``), each with: source paper, category (physics / data-driven / hybrid), and whether its parameters are platform-bound.

4.2 Stage 2 — Zero-fit transfer benchmark

For the three models that can be evaluated **as-published** (momentum theory with M100 physical constants; Tseng's reported simplified regression), compute predictions with **no fitting**. Result: energy ARE 53–76% — all fail. The zero-fit failures are not "the models are bad"; they are **platform transfer failure**. This motivates Stage 3.

4.3 Stage 3 — Refit benchmark and starting point

Refit every form's coefficients on the frozen training split. Result: data-driven polynomial (Tseng 2022) 1.90% best; physics forms 6.7–7.0%; the refit best becomes the loop's start (vs. the naive "start from textbook" that earlier pipelines used). The zero-fit-vs-refit contrast (53–76% $\rightarrow$ 1.9–7.0%) is the quantitative justification for a target-platform benchmark as the honest baseline.

4.4 Stage 4 — Cheat-proof discovery loop

A frozen held-out evaluator; an LLM proposes a featurize change each round; sandbox execution; keep/revert on search+val guards (KEEP iff ``search_ARE < best - 1e-4`` and ``val_ARE  $\leq$  best + 0.003``); a forced novelty gate (literature check before anything is called new); causal ablation (remove a term $\rightarrow$ does the planning decision revert?); honest negative reporting (negative results logged, not deleted). Convergence $\sim 1.9\%$ matches the human expert (Tseng) on the same data.

4.5 Stage 5 — Capability-boundary quantification and the Headroom Diagnostic

Statistical. Loop vs. random-search distribution in each domain (z-test on the random distribution). Energy: $z = -0.35$ (tie); planning: $z = -2.38$ (0% of random runs beat the loop).

Controlled. Fair complexity sweep: same action space (subset sampling), guided = utility-weighted, random = uniform. Result: advantage ≈ 0 at all nominal-space sizes $K = 8 \dots 200$ in energy — nominal space inflation does not help.

Algorithm 1 — Headroom Diagnostic (pre-loop prediction).

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Input: candidate pool  $P$ , frozen evaluator  $E$  (search split only), budget  $B$

1. For each term  $t \in P$ :  $u_t \leftarrow E(\text{score of } \{1, t\})$  # single-term utility
2.  $\text{spread} \leftarrow \text{std}(u)$ ;  $\text{best\_single} \leftarrow \min(u)$
3.  $\text{best\_comb} \leftarrow$  median over 5 seeds of best  $B$  random subsets
4.  $\text{synergy} \leftarrow \text{best\_single} - \text{best\_comb}$
5.  $\text{rel\_spread} \leftarrow \text{spread}/\text{best\_single}$ ;  $\text{headroom} \leftarrow \max(0, -\text{synergy})/\text{best\_single}$
6.  $\text{predict} \leftarrow \text{"tie"}$  if  $(\text{rel\_spread} > 0.3 \wedge \text{headroom} < 0.05)$  else  $\text{"guided-wins"}$

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Correctness. On 9 fine-grid K levels (8–200) in energy, the diagnostic predicts "tie" everywhere, matching the fair sweep's observed advantage ≈ 0 . In the planning domain, the diagnostic predicts "tie" within the config space — and the decomposition confirms it: config-only loop (3264) $\approx$ random (3135), while the actual win (2882) is entirely from the evolved smoother **code** (a –382-point cross-space escape). **Guided search helps iff there is an escapable higher-complexity space; the diagnostic identifies when the current space is saturated.**

Scope and limit of the diagnostic. The diagnostic answers "is the *current* space saturated?" — it does not, by itself, answer "does an escapable higher-complexity space exist?" In the energy domain, all 9 controlled settings are saturated (single-term utilities dominate; combination adds nothing); the diagnostic correctly predicts tie. In the planning domain, the diagnostic correctly predicts tie *within* the config space, and the cross-space escape (evolved code) is detected by a separate check — whether the search operates in a space where program-level changes are possible.

Mechanistic validation in a third (synthetic, controlled) search domain. To test whether the diagnostic's rule is a mechanism rather than an artifact of our two domains, we construct an independent feature-subset-selection task with controlled effective dimension (``headroom_third_domain.py``). With a single dominating term (low effective dim, mirroring the energy domain), the diagnostic correctly predicts tie

(single-term utility spread is large, combination adds nothing: best_single 0.247 $\approx$ best_subset 0.247). With interaction-dependent signal (high effective dim, mirroring the planning domain), all single terms are weak (best_single 9.204, spread $\approx$ 0) and only combinations help — the diagnostic correctly predicts guided-wins. The rule thus generalizes beyond UAV domains: **the spread/synergy signature of the space, not the application, determines the prediction.**

Honest boundary. A synthetic-probe also confirms what the diagnostic does **not** do: with noise diluting a true interaction, single-term statistics alone cannot always separate high from low synergy pools (`headroom\_mechanism\_probe.json`). The diagnostic is a reliable **saturation detector** (correct on 2 real domains + 1 controlled synthetic + fine-grid); "does an escapable program space exist" is a separate question answered by inspecting the mutation space, not the feature pool.

5. Experiments

5.1 Zero-fit transfer vs. refit (Table 1, Fig. 2–3)

Model	as-published (zero-fit) ARE	refit ARE
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Momentum hover (physics)	72.7%	—
Momentum forward (physics)	76.3%	—
Tseng simplified (paper coeffs)	53.2%	1.90%
Abeywardena VRS (physics)	—	6.94%
Dorling 2017	—	6.98%
Morbidi 2020	—	6.79%
Stolaroff 2018	—	2.39%
Full linear LIB (12 poly)	—	1.91%
Our loop (physics+payload)	—	1.93%

*Takeaway: zero-fit transfer fails; refit recovers data-driven models to $\approx$ 1.9%; physics forms stay $\approx$ 7% even refit. "Our loop" 1.93% is the loop's recorded product (agent_log

iter6); a clean re-implementation in the knowledge base measures 2.04% — a featurize-formulation detail. Both are within the " $\approx 1.9\%$ data floor" story.*

Takeaway: zero-fit transfer fails; refit recovers data-driven models to $\approx 1.9\%$; physics forms stay $\approx 7\%$ even refit. The knowledge-base leaderboard is Fig. 2 (`\pub/模型知识库排行榜.png`); the zero-fit-vs-refit gap is Fig. 3 (`\pub/as_published_vs_refit.png`).

5.2 The cheat-proof loop (Fig. 4, Fig. 5)

The loop starts from the knowledge-base winner (Tseng, 1.90%) and converges at the data floor. Instant-power analysis (`\pub/瞬时功率误差.png`) shows why the *planning* result holds: at climb, textbook BEMT misses by -131W (-20%); our loop is within $\pm 5\%$. The safeguard ablations (`\safeguard_ablation.py`): removing the frozen ruler lets a probe game search; removing the novelty gate reports prior art (Michel 2024, Nguyen 2017) as discoveries; the local-optimum escape (4.2% basin $\rightarrow$ 1.9% via external search of "add linear payload") shows the external-search branch is load-bearing.

Live demonstration of the val-discipline (loop iter 12). From the knowledge-base start, a piecewise hover/cruise model was *better on all three held-out val seeds* ($\Delta -0.006/-0.061/-0.014$ percentage points) but *worse on the search split* (2.049% vs. 1.991%, driven by a seed-0 spike). The protocol's keep rule uses search as the selector and val only as a guard, so the model was **rejected** despite its better val. This is the anti-val-overfitting discipline in action: using val for selection would leak it into training, making the held-out signal worthless. A naive pipeline would have kept the piecewise model on the strength of its val numbers — and thereby overfit to a specific val split.

5.3 Capability law and the Headroom Diagnostic (Fig. 1, Fig. 6–8)

Two real endpoints: energy tie ($z=-0.35$) and planning win ($z=-2.38$, 8.1%). One controlled negative control: nominal-space inflation ($K=8\dots 200$) yields advantage ≈ 0 . One decomposition: planning win = cross-space escape (-382 from code). One synthetic third domain: low/high effective-dimension, mechanism confirmed. The diagnostic predicts all of it before the loop runs. Fig. 1 (`\两域显著性.png`) now embeds the negative control as an inset; Fig. 7 (`\头部空间诊断_预测vs实测.png`) shows prediction vs observation; Fig. 8 (`\头部空间诊断_三域验证.png`) summarizes all three domains.

6. Honest Capability Boundaries

1. Guided search does **not** beat random in the energy domain. That **is** the finding, not a failure.
 2. Tseng is the same dataset's human result; we match (1.90 vs. 1.93), we do not claim to beat human modeling.
 3. Instantaneous power: kinematics explain <40% of variance; per-flight energy is the stable, plan-able quantity.
 4. Zero-fit transfer numbers include scale-mismatch for Tseng's simplified form — reported as approximate.
 5. The planning win is measured on a synthetic corridor; real-outdoor validation is future work.
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7. Discussion

Why nominal space size does not predict guided advantage. Bergstra and the NAS results suggest the lever is **effective** dimensionality. Our controlled sweep confirms it directly: inflating the nominal pool from 8 to 200 terms did not make guided search pull ahead, because the real signal concentrates in a handful of terms. The headroom diagnostic operationalizes this — it reads utility spread, not pool size.

Cross-space escape as the real lever. The planning win is the counterpoint: within a saturated config space the loop ties random ($3264 \approx 3135$), but the LLM can **write code** (a smoother) that random config search cannot reach. This is the measured, quantitative version of the FunSearch/AlphaEvolve program-space claim, and it is why an LLM agent — which proposes in program space — is valuable precisely when the current space is exhausted.

8. Conclusion

We presented a trustworthy autoresearch protocol for noisy engineering domains, instantiated on real UAV energy modeling and energy-aware planning, with a pre-loop headroom diagnostic that correctly identifies when the loop is

worth running. The protocol is honest by construction: it reports ties and negative results, and every safeguard is load-bearing by ablation. Its convergence matches a human expert on the same data; its capability boundaries are measured, not asserted.

Practical value of the diagnostic. Beyond prediction accuracy, the diagnostic is a **budget-saving** tool: in the energy domain it flags the space as saturated before any expensive loop runs (the loop, left to itself, would spend budget converging to the same 1.9% the knowledge base already reaches). We quantify this: the diagnostic's single-term utility computation costs $O(K)$ cheap evaluations vs. the loop's $O(\text{budget} \times \text{planning-eval})$ cost. In a saturated domain, the diagnostic saves the entire loop budget.

Measured deployment value. Following the diagnostic as a decision rule is strictly better than defaulting to running the loop:

- *Energy domain (predict = tie):* the knowledge-base start (Tseng 1.90%) *beats* the loop's own product (physics+payload 1.93% from agent_log; 2.04% in the clean zoo re-implementation). Running the loop from a good start was **net-negative** — the diagnostic's "skip" advice saves the budget *and* yields the better model.
- *Planning domain (predict = config-tie + escapable code space):* config-only loop (3264) $\approx$ random (3135), so the diagnostic's config-space "tie" is correct; the win (2882, +8.1%, $z = -2.38$) comes entirely from cross-space escape, which the diagnostic detects (a higher-complexity space exists). Investing in the loop pays.
- This is a concrete, quantitative "when to run autoresearch" rule: **skip in saturated spaces, invest where a higher-complexity space is escapable.** (``headroom_value.py`` $\rightarrow$ ``headroom_value.json``)

9. Threats to Validity & Defenses

(T1) "An LSTM beats your energy model by 72% RMSE (Muli et al.), so the energy-modeling instantiation is weak."

Defense: Our claim is **not** "best energy model." It is "trustworthy autoresearch protocol + a diagnostic for when guided search helps." The energy

model is the **instantiation domain**, deliberately chosen for its noise and lack of exact evaluator — not for maximizing ARE. We match the human polynomial baseline (1.93% vs. 1.90% per-flight energy ARE; the LSTM's 36 RMSE is per-sample, a different quantity and a black box unusable as an interpretable planning cost). We explicitly do not claim to beat black-box ML on the regression metric.

(T2) "The headroom diagnostic is only validated in 2 domains — a fitted classifier would also fit 2 points."

**Defense:* The diagnostic is a mechanism-based rule (utility spread + combination synergy → saturation), not a fitted classifier — no parameters are trained on the outcome. It is validated at 9 controlled complexity levels in the energy domain (all correctly tie) and at 2 planning settings (config tie, cross-space win). The fine-grid sweep is a systematic probe of the rule, not a fit.

(T3) "The loop is net-negative in the energy domain (1.93% > 1.90% start) — isn't the framework pointless?"

Defense:* That is precisely the point. The energy domain is a **saturated space; the honest protocol — and the diagnostic — recognize this and stop. A naive autoresearch narrative would have claimed "6.88% → 1.93% (73% improvement)" without reporting that the loop does not beat a good starting point. Our protocol's value is **knowing when not to run**, and the planning domain shows it **knows when to invest** (+8.1%).

(T4) "The zero-fit benchmark is unfair to physics models — their coefficients are platform-bound."

Defense:* That is the finding, stated explicitly: platform-bound coefficients make zero-fit transfer fail, which is why refit is a protocol stage rather than an assumption. The refit numbers (physics forms still 6.7–7.0%) then show the failure is not only coefficient-bound — the **forms also lack the flexibility of data-driven terms in this noisy regime.

(T5) "The planning domain is synthetic (a simulated corridor)."

**Defense:* Accepted as a limitation. The energy-modeling instantiation uses real DJI M100 hardware data; the planning instantiation uses the standard kinodynamic simulator in this pipeline (RotorPy + PX4 SITL validated separately). Real-outdoor planning validation is future work.

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Reproducibility

- Evaluator (frozen): ``m100_eval.py``
 - Knowledge base + leaderboard: ``energy_model_zoo.py``, ``benchmark_model_zoo.py``, ``model_zoo_leaderboard.json``
 - Zero-fit benchmark: ``as_published_models.py``, ``as_published_benchmark.py``
 - Fair complexity sweep: ``complexity_scaling_fair.py``, ``complexity_scaling_fair.json``
 - Headroom diagnostic: ``headroom_diagnostic.py``, ``planner_headroom.py``, ``headroom_finegrid.json``
 - Statistical significance: ``significance_test.py``, ``planner_significance.py`` (restored from git history)
 - Safeguard ablation: ``safeguard_ablation.py``
 - Figures: ``pubfigs.py`` + figure scripts → ``experiments/pub/``
 - All numbers trace to committed JSONs; every script is version-controlled.
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Figure Captions

Fig. 1 — Protocol overview. The four-stage trustworthy autoresearch protocol instantiated on real DJI M100 UAV energy modeling and energy-aware path planning, with measured numbers per stage: (i) a knowledge base of 14 published models; (ii) a zero-fit transfer benchmark showing all as-published models fail on the target platform (energy ARE 53-76%), motivating refit; (iii) a refit leaderboard selecting Tseng's polynomial (1.90%) as the loop's start; (iv) the cheat-proof loop; plus the quantified capability boundaries (energy tie, planning win) and the pre-loop headroom diagnostic. (pub/协议总览_Fig1.png)

Fig. 2 — Knowledge-base leaderboard. All 14 literature UAV energy models, refit on the frozen M100 evaluator (energy ARE, lower is better). Data-driven models (Tseng 1.90%, linear LIB 1.91%) dominate; all physics forms stay 6.7-7.0% even after refit. (pub/模型知识库排行榜.png)

Fig. 3 — Zero-fit vs. refit. As-published (zero-fit) models fail on the target platform (energy ARE 53-76%, predicted power far below measured 522 W), while the same forms after refit reach ~2% -- the measured justification for an explicit refit stage. (pub/as_published_vs_refit.png)

Fig. 4 — Per-state instantaneous power error. Mean predicted vs. measured power per climb-rate bin on held-out flights, for our loop, Tseng's polynomial, and textbook BEMT. At climb, BEMT misses by -13% to -20% (-131 W at steep climb) -- the mechanism behind the planning decision; our loop is within +-5% on all climb states. (pub/瞬时功率误差.png)

Fig. 5 — Safeguard ablations. Removing any one mechanism makes the loop overclaim: (A) the frozen ruler resists a gaming probe (2.13% > honest 1.99%); (B) the novelty gate demotes 2 of 3 "discoveries" to known prior art; (C) the loop's value is domain-dependent (planning win vs. energy tie). (safeguard_ablation.py + pub/框架消融_局部最优.png)

Fig. 6 — Effective-complexity law. Guided-search advantage depends on effective combinatorial complexity, not nominal space size. Two measured endpoints: energy tie ($z=-0.35$) at low effective dimension, planning win ($z=-2.38$) in code space; the controlled negative control (nominal K inflation 8->200 -> advantage ~0) is embedded as an inset. Literature markers for context. (pub/两域显著性.png + pub/复杂度规律.png)

Fig. 7 — Headroom diagnostic: prediction vs. observation. Each setting plotted at its observed advantage (pp); color = the diagnostic's pre-loop prediction (gray = tie, green = guided-wins). All predictions match observations. (pub/头部空间诊断_预测vs实测.png)

Fig. 8 — Three-domain diagnostic validation. Summary of the diagnostic across all tested settings: energy 9 K-levels (tie, gray), planning config-space (tie) vs. cross-space code escape (+8.1%, green), and a synthetic third search domain with controlled effective dimension (low -> tie, high -> guided-wins). Every prediction matches the observed outcome. (pub/头部空间诊断_三域验证.png)